Parametric Audio-Visual SELD via Variational Inference and Implicit Feature Pooling *

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Abstract

The reconstruction of dense spatial acoustic maps from 4-channel microphone arrays constitutes an underdetermined inverse problem, forcing exact numerical solvers to produce structural artifacts that convolutional detection heads misinterpret as genuine sound events. We formulate sound event localization and detection as a continuous variational inference problem over a fixed latent space of parametric Gaussians. By integrating pre-trained visual and acoustic feature fields via implicit analytical pooling, we bypass discrete spatial grids entirely. A generative Gaussian Mixture Model with a temporal prior enables variational tracking via gradient-ascent active inference, propagating the full posterior distribution across frames for artifact-free temporal coherence.

Keywords Sound Event Localization and Detection · Variational Inference · Implicit Feature Pooling · Gaussian Mixture Model · Active Inference

1 Introduction

This is a proposal for research in audio source localization and classification in real-time audio-visual settings. The task is to detect sound sources from 4 microphone and 360° camera data, and classify them into specific classes (e.g., speech, alarm, vacuum cleaner, footsteps, dog bark, baby cry, crash, scream, door knock, phone ring, laughter, cat, blender). The challenge is to make the solution fast and possible to use in real-time on small devices. The proposal is directly inspired by the DCASE challenge.

2 Variable Definitions and Feature Extraction

Let the raw input modalities be defined as follows:

- $X_v \in \mathbb{R}^{3\times H_{in}\times W_{in}}$: The raw 360-degree visual input data.
- $X_a \in \mathbb{R}^{4\times T_{in}}$: The raw 4-channel microphone audio input data.

These raw inputs are transformed via their respective neural network encoders into dense, semantically rich feature fields over the continuous spatial domain Ω:

$$H_v = f_v(X_v) \in \mathbb{R}^{C_v\times H\times W} \tag{1}$$

$$H_a = f_a(X_a) \in \mathbb{R}^{C_a\times H\times W} \tag{2}$$

where H_v represents the visual feature field, H_a represents the acoustic feature field, and C_v, C_a denote the respective channel depths.

We track a latent space of K continuous 2D Gaussians (where $K \approx 32$). The k-th Gaussian is parameterized by:

* *Citation*: **Authors. Title. Pages.... DOI:000000/11111.**

- μ_k: The spatial centroid (azimuth and elevation) tracking the location.
- Σ_k: The covariance matrix representing the spatial spread and localization uncertainty.
- $\pi_k \in [0, 1]$: The certainty that this k-th Gaussian is currently tracking an active sound source.

The localized feature vectors for the k-th source are obtained by analytically integrating the feature fields weighted by the continuous Gaussian probability density function:

$$z_{v,k} = \int_{\Omega} H_v(x) \mathcal{N}(x \mid \mu_k, \Sigma_k)\, dx \tag{3}$$

$$z_{a,k} = \int_{\Omega} H_a(x) \mathcal{N}(x \mid \mu_k, \Sigma_k)\, dx \tag{4}$$

3 Probabilistic Model and Temporal Bayesian Updating

Let Y represent the true target variables (the ground truth). The true state of the acoustic environment at a given frame consists of M discrete point sources. We define the target variable as $Y = \{(s_m, c_m)\}_{m=1}^{M}$, where $s_m \in \Omega$ is the true physical coordinate of the sound, and $c_m \in \{1, \dots, C\}$ is the true categorical class label.

Because the true point sources s_m are unknown and the sensory data is ambiguous, we model our uncertainty over the source locations using a latent parametric representation. The latent state parameters at time t are $\Theta^{(t)} = \{\mu_k^{(t)}, \Sigma_k^{(t)}, \pi_k^{(t)}\}_{k=1}^{K}$.

We frame the continuous tracking of these sound sources as a recursive Bayesian filtering problem. Let $\mathcal{D}^{(t)} = \{X_v^{(t)}, X_a^{(t)}\}$ represent the observed multimodal data at the current frame. The posterior distribution of our latent parameters is given by Bayes' theorem:

$$p(\Theta^{(t)} \mid \mathcal{D}_{1:t}) \propto p(\mathcal{D}^{(t)} \mid \Theta^{(t)})\, p(\Theta^{(t)} \mid \mathcal{D}_{1:t-1}) \tag{5}$$

The term $p(\Theta^{(t)} \mid \mathcal{D}_{1:t-1})$ is the temporal prior. We assume that the physical properties of sound sources change continuously over time, governed by a transition model $p(\Theta^{(t)} \mid \Theta^{(t-1)})$. The temporal prior for the current frame is calculated by marginalizing over the posterior distribution from the previous iteration:

$$p(\Theta^{(t)} \mid \mathcal{D}_{1:t-1}) = \int p(\Theta^{(t)} \mid \Theta^{(t-1)}) p(\Theta^{(t-1)} \mid \mathcal{D}_{1:t-1})\, d\Theta^{(t-1)} \tag{6}$$

In practice, applying this transition model effectively "blurs" or broadens the covariance of the previous posterior, reflecting the increased spatial uncertainty as time advances. This loosened prior acts as a structural regularizer during the optimization of frame t, anchoring the new Gaussians to their previous trajectories while allowing them enough flexibility to track moving subjects.

Finally, the classification is modeled independently from the spatial generative process. The predicted categorical distribution for the k-th latent source relies on the implicitly pooled visual and acoustic features, mapped through a classification head:

$$P(\hat{c}_k \mid X_v^{(t)}, X_a^{(t)}) = \text{Softmax}(W[z_{v,k}^{(t)}, z_{a,k}^{(t)}] + b) \tag{7}$$

During training, the approximate posterior parameters $\Theta^{(t)}$ are optimized to maximize the evidence of the observed data, while the class predictions $\hat{c}_k$ are optimized against the true categories c_m via cross-entropy.

4 Variational Inference and Optimization

Exact posterior inference for the above formulation is intractable. We introduce a tractable variational distribution $q^{(t)}(\Theta)$ to approximate the true posterior. The system is optimized by maximizing the Evidence Lower Bound (ELBO). The objective incorporates both the generative spatial likelihood and the discriminative class certainty derived from the implicitly pooled multimodal features:

$$\mathcal{L}(q^{(t)}) = \mathbb{E}_{q^{(t)}} \left[\log p(\mathcal{D}^{(t)} \mid \Theta^{(t)}) + \log p(C^{(t)} \mid Z^{(t)}) \right] - D_{KL} \left(q^{(t)}(\Theta) \parallel p(\Theta^{(t)} \mid \mathcal{D}_{1:t-1}) \right) \tag{8}$$

where $C^{(t)}$ represents the matched semantic class labels and $Z^{(t)}$ is the set of pooled features. The Kullback-Leibler divergence term ensures that the current spatial estimation remains anchored to the temporal prior established in the preceding frame.

4.1 Gradient-Ascent Active Inference

We optimize the ELBO iteratively during the forward pass. Initializing the current state with the previous frame's posterior, we apply M steps of gradient ascent. Let m denote the iteration index. The update equations for the spatial parameters are:

$$\mu_k^{(t,m+1)} = \mu_k^{(t,m)} + \eta_\mu \nabla_{\mu_k} \mathcal{L}(q^{(t,m)}) \tag{9}$$

$$\Sigma_k^{(t,m+1)} = \Sigma_k^{(t,m)} + \eta_\Sigma \nabla_{\Sigma_k} \mathcal{L}(q^{(t,m)}) \tag{10}$$

These updates analytically shift the latent Gaussians across the spatial domain. The gradients direct the parameters to maximize intersection with valid acoustic signals and visual class probability (such as aligning with a detected object in the visual feature field), while penalizing divergence from the temporal prior.

5 State Propagation and Prediction

Upon convergence of the active inference loop at frame t, explicit coordinates must be extracted for evaluation. We extract the Maximum a Posteriori (MAP) estimate from the fitted variational distribution:

$$\hat{\Theta}^{(t)} = \arg\max_{\Theta} q^{(t)}(\Theta) \tag{11}$$

However, the MAP estimate alone is insufficient for continuous tracking. The entire approximate posterior $q^{(t)}(\Theta)$, containing both the spatial coordinates and the covariance matrix representing localization uncertainty, is retained in memory. This full distribution is propagated to frame $t+1$ to construct the subsequent temporal prior, effectively bypassing the computational overhead of iterative grid rendering from a blank state.

6 Discussion and Rationale

The primary purpose of designing the architecture this way is to ensure the system inherently remembers things from previous iterations. By propagating the variational posterior across frames rather than relying on instantaneous point estimates, the model maintains a continuous memory of both source location and spatial uncertainty. Furthermore, theoretically, a sufficient number of Gaussians can model any arbitrary probability distribution. This mathematical property ensures that utilizing parametric continuous functions does not limit the model's capacity to represent complex spatial acoustic phenomena.

Regarding the optimization scheme and the contrast with existing baselines, the rationale for this approach is as follows:

- We tested previously generating acoustic maps directly from the microphone data.
- The minimization scheme proposed in those previous works is highly overparameterized.
- The true minimizer of that exact objective function is full of spatial artifacts.
- Authors of previous baseline works hide these structural artifacts by employing early stopping during the numerical routine.
- Treating these early-stopped, artifact-prone maps as ground truth forces standard convolutional detection heads to predict non-existent sound sources.
- We introduce the parametric Gaussians in order to make the optimization scheme much faster and way less prone to overfitting, which in our specific case means eliminating the generation of these spatial artifacts.

By explicitly defining the target space as a combination of continuous parameters rather than a discrete pixel matrix, the architecture remains computationally lightweight, strictly adheres to physical constraints, and supports real-time deployment requirements.

Acknowledgments

References